

Adithya Krishnan K S

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LinkedIn — GitHub

Education

Cochin University of Science and Technology

B.Tech in Information Technology

2024 – Present
Semester 5 — CGPA: 10.0/10.0

- S1 GPA: 10.0 S2 GPA: 10.0 S3 GPA: 10.0

Don Bosco Higher Secondary School

Higher Secondary Education – Bio-Maths

2019 – 2021
Irinjalakuda, Kerala, India

- Secured 99.6% in Higher Secondary Examination.

Experience

IIIT Kottayam

Research Intern

June 2026
Kerala, India

- Conducted research on **incremental concept drift detection and noise-resistant data stream analysis**.
- Implemented and evaluated **NR-EIDDM and DenStream-based drift detection approaches** using Python and streaming-data techniques.
- Conducted experiments on benchmark datasets and analyzed **precision, recall, F1-score, and detection delay** to investigate improvements in noise handling and adaptive drift detection.

MathWorks

Computational Thinking & Artificial Intelligence Intern

June 2025
Remote

- Applied AI and computational thinking workflows including data preparation, statistical analysis, model development, simulation, and evaluation using MATLAB.
- Built predictive and algorithmic models leveraging machine learning, optimization techniques, signal processing, and data visualization to analyze system behavior.
- Modeled and validated dynamic systems and circuits in Simulink using block-diagram modeling, control logic, and simulation outputs.

CitySense Urban Intelligence Startup

Managing Partner

2025 – Present
Kerala, India

- Leading an IoT-based smart-city startup focused on flood detection, mosquito surveillance, and urban monitoring solutions.
- Rebuilt a patented flood detector prototype by integrating sensor networks and real-time data analytics for urban risk assessment; project funded by RUSA and approved by Kochi Municipality.
- Overseeing R&D on AI-driven rainfall prediction models and scalable smart-city analytics platforms.

Projects

Dayflow HRMS

Jan 2026

React, Supabase, PostgreSQL, RBAC

- Designed and developed a full-stack HRMS to manage attendance, leave requests, payroll workflows, and employee profiles using role-specific dashboards.
- Implemented secure authentication and role-based access control using Supabase Auth and Row-Level Security (RLS).
- Built real-time attendance tracking with check-in/check-out functionality and historical activity views using Supabase Realtime.
- Designed PostgreSQL schemas, SQL triggers, and backend logic to automate employee onboarding and leave approval workflows.

UniRewards

Apr 2026

Node.js, Express.js, PostgreSQL/Supabase, Firebase

- Developed a full-stack university reward marketplace where students earn points from faculty, transfer points to peers, and redeem rewards through a competitive auction-based marketplace.
- Implemented role-based workflows for Students, Professors, and Admins, including dynamic faculty point allocation using configurable **Allocation Rules** and centralized administrative controls.
- Built a hybrid backend architecture using **Node.js, Express.js, PostgreSQL/Supabase, and Firebase Firestore**, combining relational transaction management with flexible rule storage.
- Ensured secure and reliable transactions using **JWT authentication, RBAC, RLS, and PostgreSQL RPC/stored procedures** to maintain point integrity and prevent race conditions during transfers and bidding.

Flood Prediction System using Hybrid LSTM Model

Nov 2025

Python, TensorFlow, Keras

- Designed a hybrid deep learning model combining LSTM-based time-series sensor data with meteorological features for flood prediction.

- Engineered multivariate temporal datasets using sliding windows, MinMax normalization, and feature scaling.
- Built a multi-input neural network architecture and achieved 100% validation accuracy on the test data.
- Converted the trained model to TensorFlow Lite (TFLite) for edge deployment and visualized predictions using Matplotlib.

Smart Hospital Management System

React, Node.js, PostgreSQL, Supabase

Dec 2025

- Developed a hospital appointment and queue management system for patient registration, booking, and live doctor queues.
- Built a React + Vite frontend with role-based dashboards and live token tracking.
- Implemented a Node.js and Express REST API with priority-based queue logic and wait-time estimation.
- Integrated Supabase for OTP-based authentication and real-time queue updates.

Technical Skills

Languages: C++, JavaScript, Python

Machine Learning: Feature Engineering, Data Preprocessing, Model Evaluation

Frameworks & Libraries: TensorFlow, Keras, Pandas, Scikit-learn

Frontend: React.js, HTML5, CSS3, Tailwind CSS

Backend: Node.js, Express.js, Flask, PHP

Databases: PostgreSQL (Supabase), MongoDB, SQL

Tools & Hardware: Git, GitHub, MATLAB, Simulink, Arduino, NodeMCU

Certifications

- Complete Full Stack Web Development Bootcamp – Udemy
- Getting Started with Generative AI – IBM SkillsBuild
- Data Science and Analytics – HP LIFE